Final Project: Rubric

Statistical Evaluation of AI · Spring 2027

The project is worth 100 points, which make up 40% of the course grade. The *Final Project: Description* says what each deliverable contains. This rubric says how each one is graded.

Deliverable	Due	Points
Milestone 1: Introduction and Methods	Tue Mar 23	15
Milestone 2: Results	Tue Apr 6	20
Milestone 3: Discussion and storyboard (completion)	Tue Apr 20	5
Final presentation: slides and video	Sun Apr 25, 11:59 pm	30
Oral defense	Tue Apr 27 or Thu Apr 29	20
Peer assessment	Week 15	10
Total		**100**

The milestones and the final presentation receive one score per team. The oral is scored for the team, but a member who clearly cannot explain the team's work may be scored separately. The peer assessment is individual.

How points are given. Each row below describes full credit. Work with a minor gap loses about a quarter of the row's points; work with major gaps earns about half or less; missing work earns zero.

The writing standard

Several rows below refer to this standard. It applies to the milestones, the slides, and the narration. Writing meets it when all four of these hold.

- **Specific.** Claims carry their numbers, their cases, and the client's terms. No sentence could be pasted into another team's project unchanged.
- **Plain.** Words say what was done ("we compared," "we re-ran") rather than what sounds thorough. See the word table in the Description, Section 4.
- **Consistent.** Each thing has one name, and each number matches everywhere it appears.
- **In sentences, in your voice.** Arguments are made in sentences; bullets are only for lists of parallel items. The narration sounds like your team explaining its work, not a script being read.

The rubric grades these qualities, not which tools produced a draft.

Milestone 1: Introduction and Methods (15 points)

Criterion	Points	Full credit
Introduction	4	The client and decision are explicit. Each of two to four questions names the decision it informs and the estimand that answers it. The client's population and the benchmark's population are both stated and compared.
Data and exploratory analysis	4	A few purposeful figures, tables, and inspected cases, each with a finding that shapes or qualifies the plan. Covers balance, where the labels come from and how good they are, hard cases, and how the benchmark differs from the client's population, including which way that difference would push the results.
Evaluation design	4	The floor and the fitted baseline are named, with what the baseline uses and ignores. The baseline and AI system share the same unit, split, metric, and target, paired when they see the same cases. The uncertainty method fits the dependence in the data. The holdout is untouched. The assumption most likely to threaten the conclusion is named.
Writing	2	Meets the writing standard. Reads as the first two sections of a report, in paragraphs, that someone outside the course could follow.
Reproducibility and AI use	1	The linked notebook reproduces every number. The AI-use paragraph says what AI was used for, what was checked, and one thing it got wrong.

Milestone 2: Results (20 points)

Criterion	Points	Full credit
Revision	2	The "what changed" list responds to every piece of Milestone 1 feedback. Methods now includes the full sensitivity plan, with the result that would change the recommendation, stated before the analysis is run.
Fitted baseline	5	Reported with uncertainty and clearly distinct from the floor. The team explains what it uses, what it ignores, how it fails on real cases, and why it is a fair comparison.
Comparison with uncertainty	5	The baseline and AI system are compared on the same cases, paired where possible, with intervals that respect the dependence in the data. Both statistical and practical importance are addressed.
Calibration and errors	3	Calibration or confidence evidence when the decision uses scores, or a sentence on why it does not. Shows where errors concentrate, including at least one meaningful subgroup with its denominator.
Results as answers	3	The Results answer the Introduction's questions in order. Each paragraph opens with a sentence stating its finding, and each figure has a one-sentence takeaway in the client's terms.
Writing and reproducibility	2	Meets the writing standard. The notebook reproduces every number, and the AI-use paragraph is updated.

Milestone 3: Discussion and storyboard (5 points, completion)

Rough work is fine. Each item earns its points if it is present and complete.

Item	Points	Complete when
Sensitivity result	1	Run as planned, with the pre-stated rule honored. Says whether the conclusion held.
Discussion draft	1	Gives the recommendation and its conditions, the largest limitation, the result that would change the recommendation, and the next data to collect.

Item	Points	Complete when
Storyboard	2	Written by the team in class on Apr 20 without AI. Every slide title in order as a full-sentence claim, grouped by IMRaD section, with the figure or table each slide shows and who presents it.
Revision example	1	One AI-drafted passage next to the team's revision, with a sentence or two on what changed and why.

Final presentation: slides and video (30 points)

Criterion	Points	Full credit
IMRaD structure	6	The Introduction gives the client, the decision, the question, and a one-sentence preview of the answer. Methods gives enough about the data, baseline, AI system, and design to trust the results. Results answer the question in order, with uncertainty. The Discussion gives the recommendation, its conditions, the main limitation, and what would change it. Time is spent roughly as the Description suggests.
One story	8	The slide titles alone tell the argument, recommendation included. Every result shown is used in the Discussion, and nothing is shown that is not. The Results answer the question the Introduction asked. Names and numbers match across the slides, the narration, and Milestone 3. Speakers hand off with transitions rather than starting over.
Statistical evidence	10	The baseline and AI comparison, its uncertainty, calibration where relevant, and the sensitivity result are correct and interpreted for the client. No claim is stronger than its evidence.
Clarity and voice	4	The slides and narration meet the writing standard. Slide text is short and specific, and the narration is in the team's own words.
Figures and format	2	Figures are readable at video size, with labeled axes and units, uncertainty shown, and one message each. The video is five minutes or less, all four members speak, and the appendix includes the AI-use slide.

For **One story**, a typical minor gap is one unused result or one mismatched number. A major gap is a sequence of results with no thread connecting them to the decision.

Oral defense (20 points)

Criterion	Points	Full credit
Understanding	7	Members explain the team's choices, methods, and conclusions in their own words, including parts they did not lead.
Statistical defense	7	Members justify the baseline, uncertainty, calibration, and sensitivity choices, and connect each result to the recommendation.
Responsiveness	4	Answers to follow-up questions are honest. Members revise a claim when the evidence is weaker than they said.
Shared ownership	2	Understanding is spread across the team rather than concentrated in one person.

Peer assessment (10 points, individual)

Criterion	Points	Full credit
Your form	2	Submitted on time. Rates yourself and every teammate, with specific examples.
Your contribution	8	Consistent, appropriate contribution, judged from your teammates' forms, the milestone record, and the oral. Documented, substantial non-contribution lowers this score and can lower your other project points.

What earns credit

A simple analysis that the team can explain and defend beats a complicated one it cannot. A baseline that beats the AI system, or a conclusion of "we can't tell yet" backed by the uncertainty, can earn full credit. What loses credit is a claim stronger than its evidence.